User Input

Could you help me to annotate marker gene from this single-cell RNA-Seq data? Please list all maker genes and produce figures for all analysis.

h5ad

Analysis Workflow Steps

Step 1: Data Reading

Step 2: Quality Control (QC)

Step 3: Normalization and Highly Variable Gene Selection

...

Step 6: Marker Gene Identification and Visualization

BioMaster

Plan Phase

Plan RAG

bioRxiv, arXiv, PubMed

Plan Agent

Generate plans to achieve goals

Support Tools

BWA-MEM
HTseq
HISAT2
DESeq2
MACS3
HOMER
BCFtools
samtools
FastQC
Trimmmomatic
GATK
...

Execute Phase

Execute RAG

Read the Docs

Tool Manager

R, Python, CONDA

Task Agent

Generate tool-based scripts

Use Tool

Debug Agent

Detect and correct errors

Success

Error

Check Agent

Validate outputs and amend plan

Validation Failed

Final Output

Summary History

memory module

$\times N$

Result





